



PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Fundamentals of ICT Lab 07	Instructor: Ms.Quratulain
Program: AI Fall 2025	Date: 22th November, 2025
Marks: 10	Lab Engineer: Ms. Sibgha Noor
Student Name: Eshal Ayub	Reg. No: B25F1061AI100
Instructions: • Submit the file with your names following your registration numbers like AI001_Name. • Copy Paste work will be given no marks.	

LAB TASK

Lab Learning Objectives:

1. To learn how to create and design tables in Microsoft Access with correct fields and data types.
2. To understand how to enter and manage data in a table using Datasheet View.
3. To practice creating queries to filter, sort, and retrieve specific information from a database.
4. To apply logical conditions (like >, <, AND) in queries for precise data selection.
5. To develop skills in organizing and displaying database information effectively.

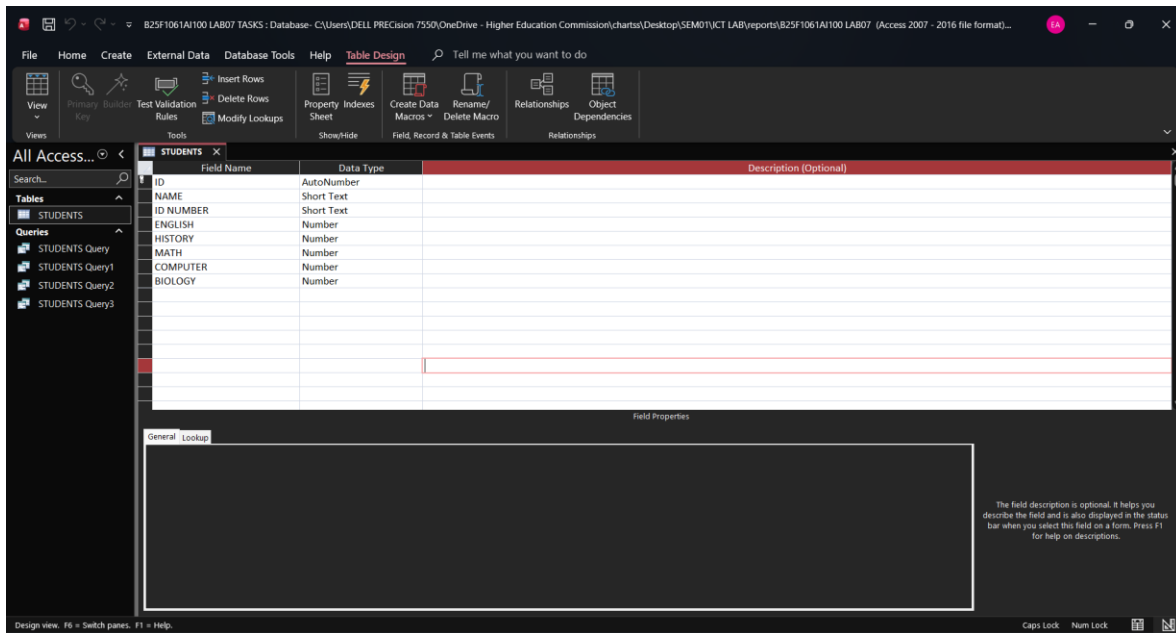
Task 1: Creating the Students Table

Explanation:

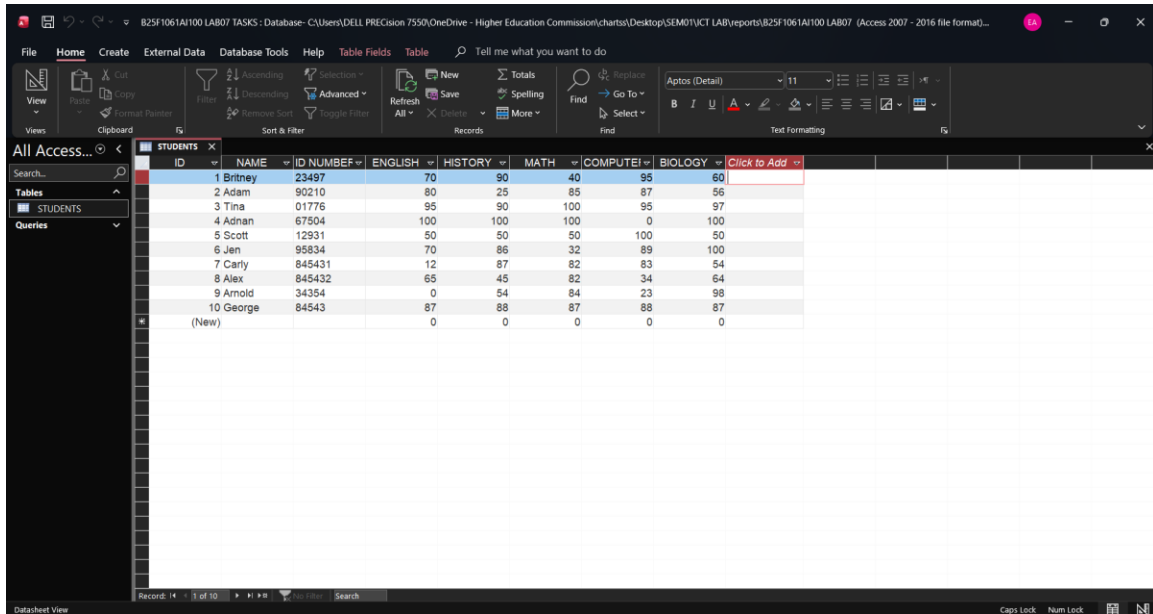
In this lab task, I created a new database in Microsoft Access and designed a table for storing students' marks. I first opened Access and selected "Blank Desktop Database," then saved the file with an appropriate name. After the database was created, I switched the default table into Design View so I could define all the fields.

I added fields such as Name, ID number, English, History, Math, Computer, and Biology, and selected suitable data types for each one. All marks were set to "Number," while Name and ID were set to "Short Text." After defining the structure, I saved the table and switched to Datasheet View. Finally, I entered all the student records provided in the lab manual one by one. This completed the setting up of the student's table.

Screenshot (Design View):



Screenshot (Datasheet View):



ID	NAME	ID NUMBER	ENGLISH	HISTORY	MATH	COMPUTER	BIOLOGY
1	Britney	23497	70	90	40	95	60
2	Adam	90210	80	25	85	87	56
3	Tina	01776	95	90	100	95	97
4	Adnan	67504	100	100	100	0	100
5	Scott	12931	50	50	50	100	50
6	Jen	95834	70	86	32	89	100
7	Carly	845431	12	87	82	83	54
8	Alex	845432	65	45	82	34	64
9	Arnold	34354	0	54	84	23	98
10	George	84543	87	88	87	88	87
(New)			0	0	0	0	0

Task 2: Queries

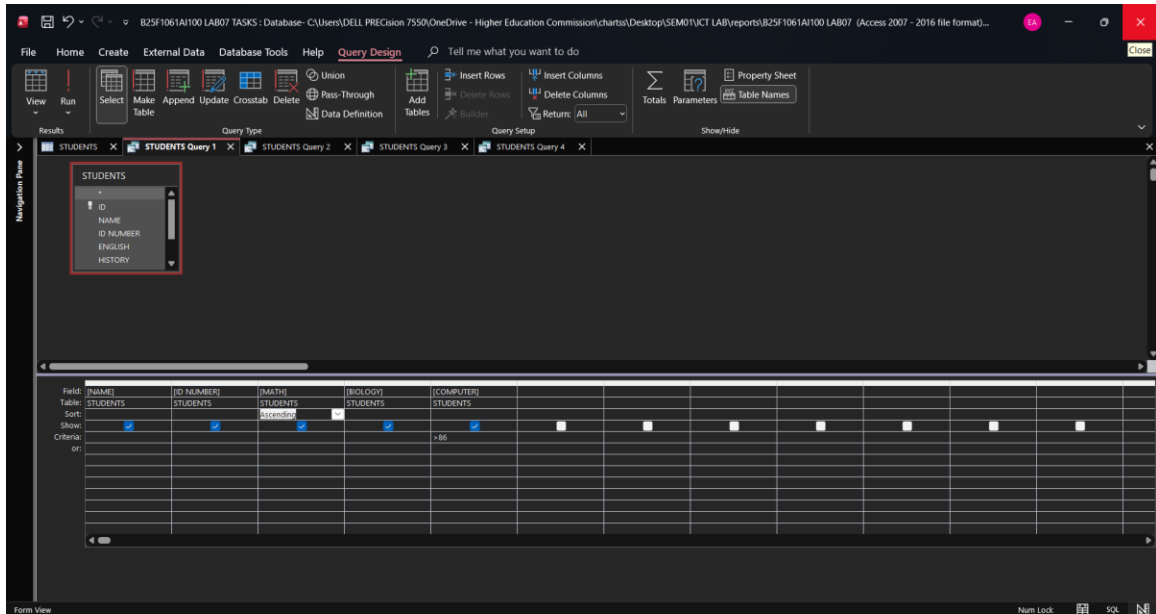
Query 1:

Explanation:

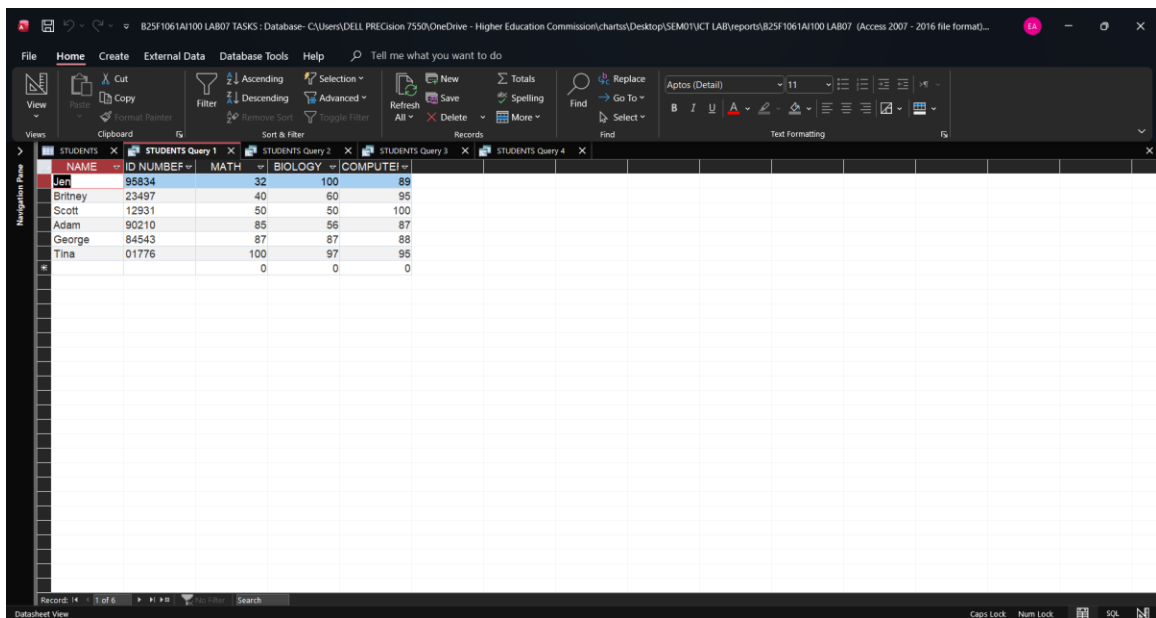
In this query, I had to find students whose Computer marks were greater than 86 and then display only their Name and ID. To do this, I opened Query Design and added the Students table. I selected the fields Name, IDNum, Math, Biology, and Computer.

In the Computer column, I typed >86 in the Criteria row to filter out students with lower marks. Since the question asked to sort by Math marks, I applied “Ascending” sorting under the Math field. After running the query, Access displayed only the students who met the condition, sorted from lowest to highest Math marks.

Screenshot (Design View):



Screenshot (Datasheet View):



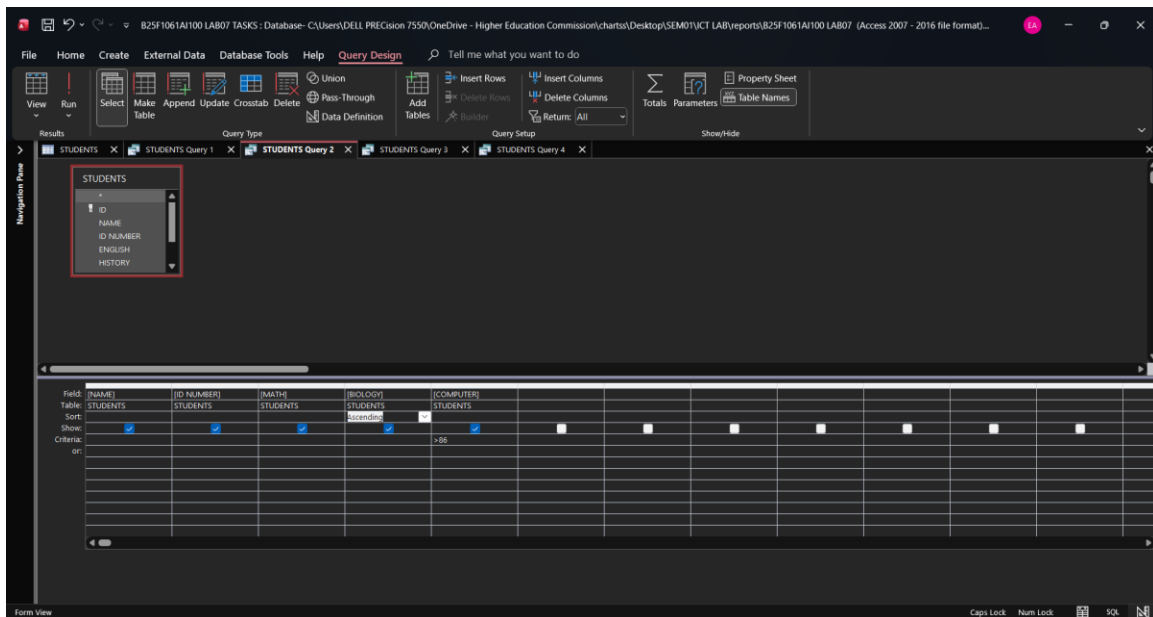
Query 2:

Explanation:

This query was like the first one, except the sorting had to be done on Biology instead of Math. I opened a new Query Design window and added the same required fields — Name, IDNum, Math, Biology, and Computer.

Again, I placed >86 in the computer field's Criteria row. This time, I changed the sorting in the biology column to "Ascending." When I ran the query, Access showed all students with Computer marks above 86, arranged according to their biology results.

Insert Screenshot (Design View):



Insert Screenshot (Datasheet View):

NAME	IDNUMBER	MATH	BIOLOGY	COMPUTER
Scott	12931	50	50	100
Adam	90210	85	56	87
Britney	23497	40	60	95
George	84543	87	87	88
Tina	01776	100	97	95
Jen	95834	32	100	89
		0	0	0

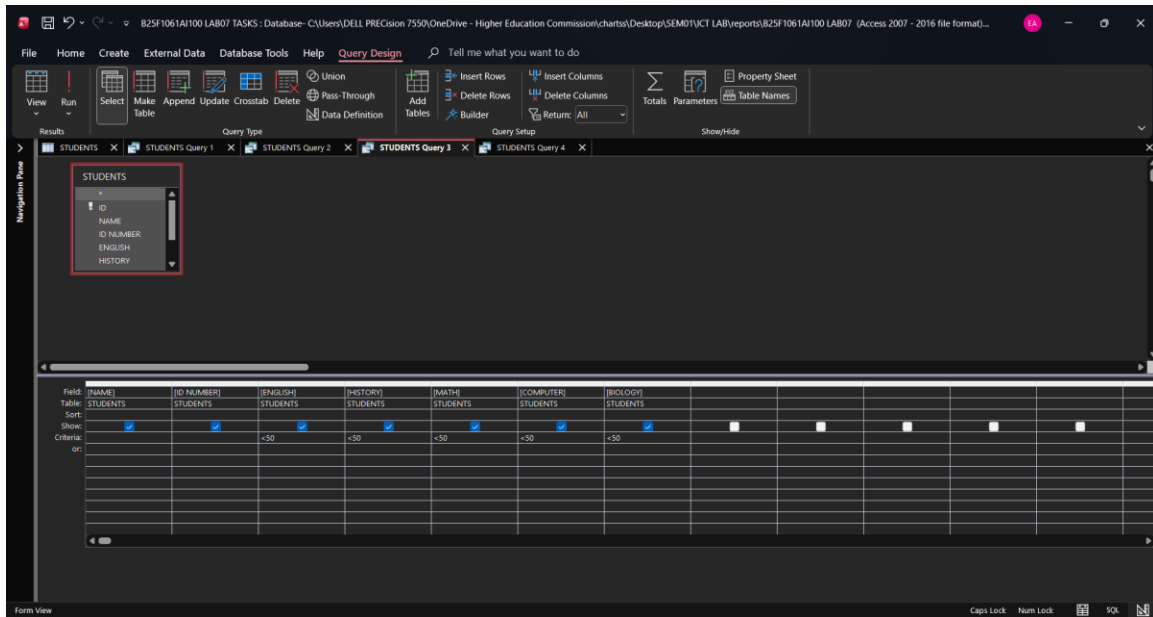
Query 3

Explanation:

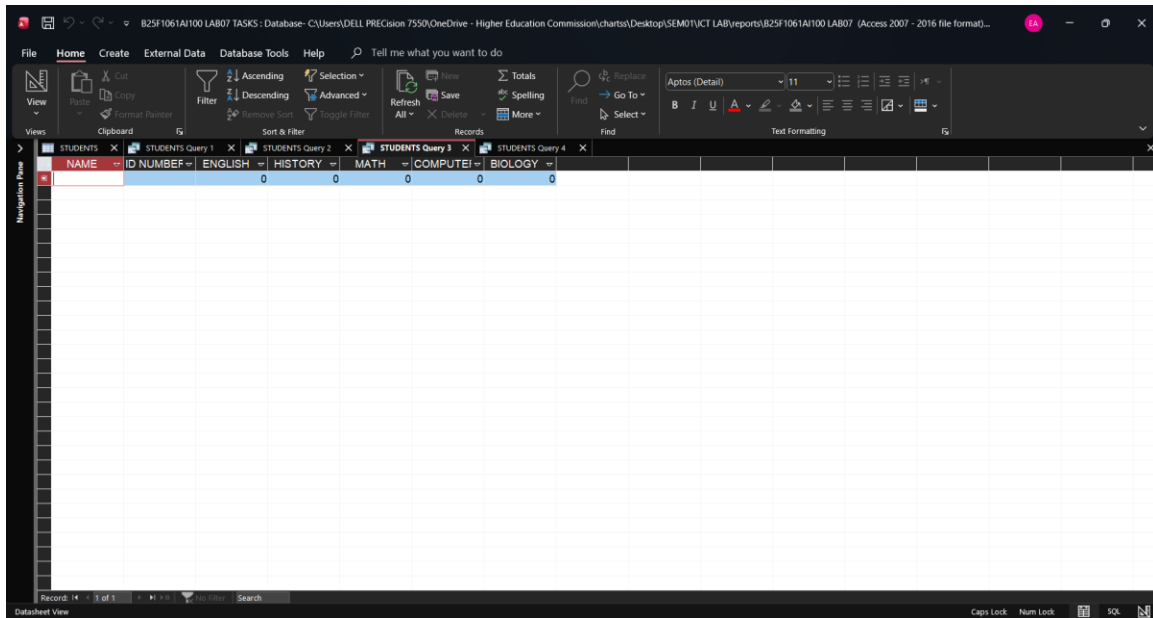
In this task, I had to find students who scored below 50 in **every** subject. I created a new query and selected all fields: Name, IDNum, English, History, Math, Computer, and Biology.

To filter correctly, I entered <50 under each subject field in the same criteria row. This forces Access to only show students who meet all conditions at once. After running the query, the result showed whether any students matched the criteria. This step helped me understand how AND conditions work in Access queries.

Insert Screenshot (Design View):



Insert Screenshot (Datasheet View):



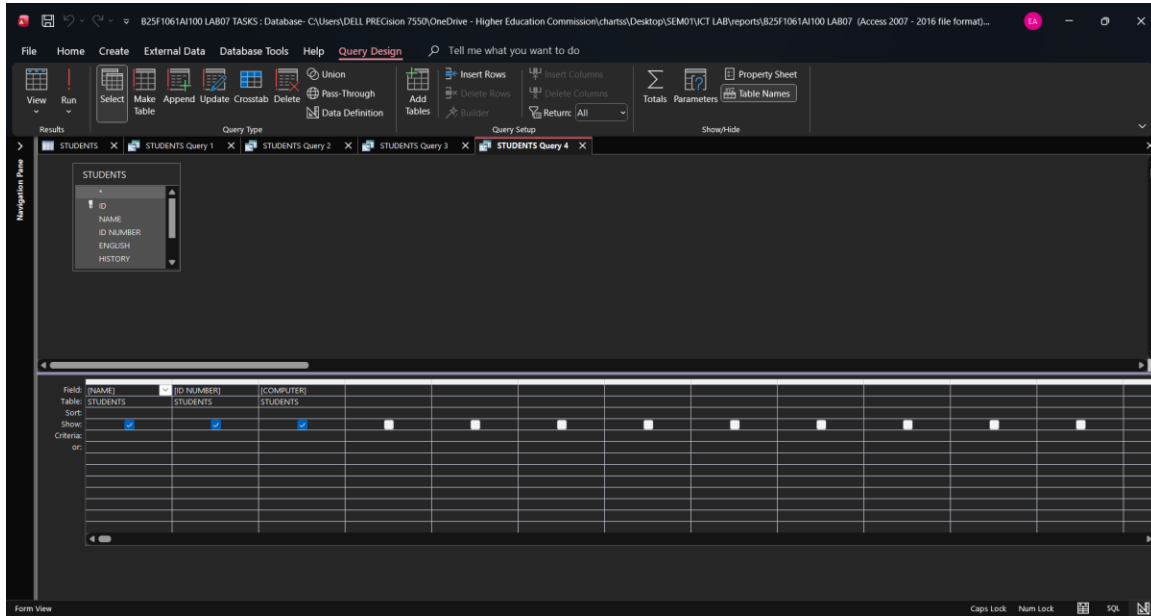
Query 4:

Explanation:

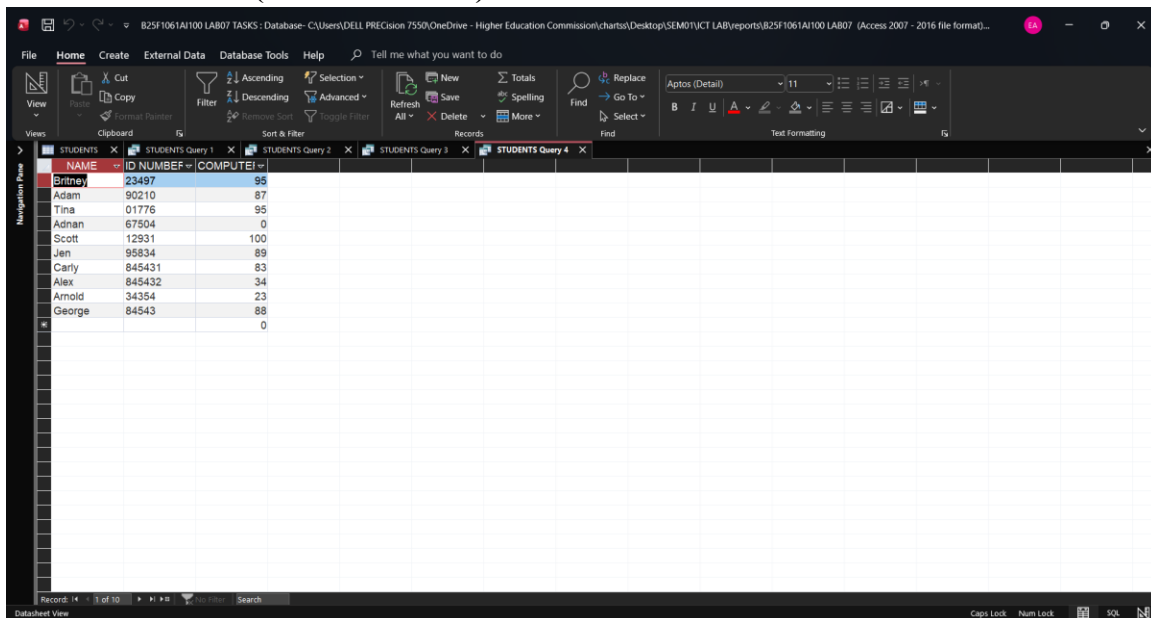
The goal of this query was to list all students enrolled in the Computer course. Since every student has a value in the Computer field, no special filtering was required.

I opened a new query, added the Students table, and selected Name, IDNum, and Computer. Because there was no criteria needed, I simply ran the query. Access displayed all students from the table. This task helped me practice selecting specific fields without applying filters.

Insert Screenshot (Design View):



Insert Screenshot (Datasheet View):



Conclusion

In this lab, I successfully created a student's table in Microsoft Access, defined appropriate fields, and entered student data. I also practiced writing queries to filter and sort information based on different criteria, such as finding students with high marks or low scores in all subjects. This exercise helped me understand how tables and queries work together in Access, and how to use conditions to extract meaningful information from a database. Overall, the lab strengthened my ability to manage and analyze data efficiently.